

BIO-INSPIRED COMPUTING

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by

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BIO-INSPIRED COMPUTING

ABSTRACT

Bio-inspired computing applies concepts and mechanisms found in nature to solve complex computational problems. This report surveys and analyses two broad sets of research. It bridges natural sciences and computer engineering. Bio-Inspired Computing across three application domains - bio-inspired quantum computing, bio-inspired building thermal designs, and bio-inspired cloud task scheduling. Genetic Algorithm (GA) - a classic optimization method based on natural selection and evolutionary principles. These studies demonstrate how biological processes such as photosynthetic energy transfer, insect sensory systems, animal group behaviour and evolutionary genetics can inspire computing methods that are more energy-efficient, adaptive and scalable. The conclusions show that nature's strategies offer robust solutions to engineering and computational challenges, from disease detection in deep learning to UAV path planning. Each in that shows how natural processes such as evolution, insect sensory abilities and the collective behaviour of animals and birds can be translated into algorithms or design principles. The studies demonstrate that learning from biology leads to computing systems that are more adaptive, energy-efficient and capable of solving problems that are very hard for traditional methods. Unlike conventional computing which relies only on human-designed algorithms, bio-inspired methods emerge from observing natural systems. These include evolutionary processes, neural networks, and swarm intelligence all of which have evolved over millions of years to solve complex tasks efficiently. It emphasises the transformative potential of mimicking nature to overcome current technological limitations.

Keywords: Genetic Algorithm (GA), robust solutions, evolutionary processes, neural networks, swarm intelligence, transformative potential

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1. INTRODUCTION

Bio-Inspired Computing is a multidisciplinary approach that borrows principles from biology - self-organization, adaptation, evolution and swarm behaviour - and translates them into algorithms, hardware designs and intelligent systems.

Quantum Computing Inspired by Chloroplasts: Photosynthetic processes show stable quantum behaviour at room temperature, suggesting new designs for quantum processors that do not require extreme cooling.

Bio-Inspired Building Facades: The silkworm cocoon's layered structure informs energy efficient building skins and can also guide thermal management in computing hardware.

Cloud Task Scheduling Inspired by Lyrebirds: Animal escape and hiding behaviours motivate new optimization algorithms (e.g., Lyrebird Falcon Optimization) for efficient cloud resource scheduling.

Bio-inspired computing takes concepts from biology - for example natural selection, self-organisation, swarm intelligence and animal sensory systems - and applies them to computing, engineering and data analysis. Instead of programming every detail, we imitate how living organisms survive, adapt and optimise. Nature has evolved powerful strategies for survival and optimization; computing can adapt these strategies for Scalability in large search spaces, Adaptability to dynamic environments.

2. GENETIC ALGORITHM

Genetic Algorithm (GA): A well-known bio-inspired optimization technique that simulates natural selection to find near-optimal solutions to large complex problems. They are used to find optimal or near-optimal solutions to complex problems, especially when:

- There's no exact solution or it's hard to compute directly.
- The search space is large (too many possibilities).

2.1 Common Use Cases

- Feature selection in Machine Learning
- Route optimization (e.g. travelling salesman problem)
- Scheduling (like job timetables)

- Game strategy development
- Engineering design problems

2.2 Pseudo Code

Population:

A group of possible routes through all cities.

More variety (diversity) → better chance of finding a short route.

Fitness Evaluation:

Measures how good a route is by its total distance.

Shorter distance → higher fitness.

Selection:

Chooses which routes become parents.

Better routes are more likely to pass their “plan” to the next generation.

Crossover (Recombination):

Combines two parent routes to make a new route.

Keeps city order and avoids duplicates.

Mixes good parts of both parents.

Mutation:

Randomly swaps cities in a route.

Introduces new ideas and prevents routes from being too similar.

2-Opt Local Improvement:

Checks pairs of edges and reverses them if it makes the route shorter.

Quickly improves routes without randomness.

Elitism:

Always keeps the best routes from one generation to the next.

Ensures GA never loses the best solution found so far.

Iteration (Generations):

Repeat selection, crossover, mutation, and 2-opt many times (routes become shorter)

3. QUANTUM COMPUTING INSPIRED BY CHLOROPLASTS:

3.1 Structure and Components

Quantum computers are advanced machines that leverage the principles of quantum mechanics to perform computations far beyond the capacity of classical computers. At the core of these systems are qubits, the basic units of quantum information. Unlike classical bits that exist only as 0 or 1, qubits can exist in superpositions of states, allowing them to represent multiple values simultaneously. These qubits are typically implemented using superconducting circuits, trapped ions, or photons, each method requiring precise engineering to function effectively. To preserve the delicate quantum states, quantum computers depend on cooling systems that bring the hardware close to absolute zero, drastically reducing thermal noise and preventing qubits from collapsing into classical states. In addition to qubits and cooling, quantum computers utilize quantum gates and circuits, which manipulate qubit states and enable entanglement, interference, and other key quantum operations that make complex computations possible. Since quantum systems are inherently prone to noise and decoherence, error correction units are critical for maintaining accuracy, ensuring that calculations remain reliable despite environmental disturbances.

3.2 Working of a Quantum Computer (Inspired by chloroplasts)

Challenges in Conventional System: Conventional quantum computers face persistent problems of instability, high energy consumption, and the requirement for near-absolute-zero temperatures to prevent decoherence.

Exciton Coherence Inspiration: By studying chloroplasts, researchers propose a bio-inspired working model of quantum computing. The concept of exciton coherence inspires the design of qubits that can maintain their quantum state without extreme cooling, making the system inherently more stable and energy-efficient.

Thylakoid-Inspired Organization: The layered thylakoid arrangement provides a natural guideline for organizing qubits spatially, reducing interference among them and creating pathways that optimize interactions, much like how energy flows within chloroplast membranes.

Natural Error Tolerance: The error-tolerant energy transfer pathways in plants suggest strategies that can help quantum systems minimize decoherence and noise, reducing dependence on artificial error correction.

Overall Reimagining: This bio-inspired approach reimagines quantum computing as a system that stabilizes itself through natural design principles rather than relying exclusively on cooling infrastructure and external correction mechanisms.

3.3 Structure of Quantum Computing

Bio-Inspired Qubit Layout: The proposed structure translates chloroplast principles directly into quantum hardware. The qubit layout is modelled after the layered, elliptical structure of thylakoid membranes, creating an arrangement that promotes stability and coherence by mimicking photosynthetic efficiency.

Efficient Spatial Organization: This natural template ensures that qubits are organized in a way that maximizes their interaction while minimizing disruptive interference, enhancing overall system stability.

Energy Transfer Mechanism: The photosynthetic process, where excitons travel seamlessly across networks, inspires a quantum system in which qubits can communicate more effectively. This design reduces decoherence and lowers the overall energy cost of operations.

Practical Implications: Such a system paves the way for quantum processors that can function under less restrictive environmental conditions, moving toward scalable, efficient and practical deployment.

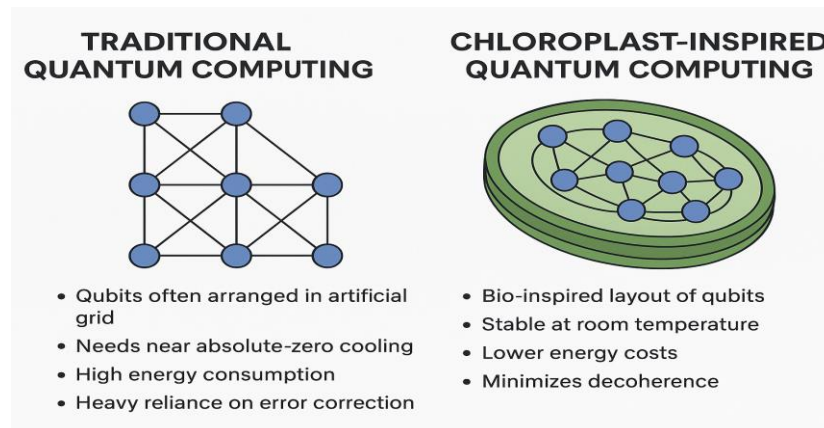


Figure 1 Quantum Computing Inspired by Chloroplasts

3.4 Outcome

The outcome of this bio-inspired approach is a vision of scalable and stable quantum processors that address the major limitations of current quantum technologies. By reducing dependency on extreme cooling systems, these designs significantly lower operational costs and energy demands. The stability achieved through bio-inspired layouts and coherence mechanisms not only makes quantum computing more practical but also enhances its potential for large-scale deployment in real-world applications. This paradigm shift, guided by natural models like chloroplasts, suggests a pathway toward the next generation of quantum computers machines that combine the principles of physics with the proven strategies of biological systems to achieve unprecedented efficiency, reliability, and scalability

4. MULTI-SKIN FACADE BUILDINGS INSPIRED BY SILWORM COCOONS

4.1 Structure and Components

It explores the structure and components of a bio-inspired multi-skin façade system modeled after the thermal-regulating properties of silkworm cocoons. This façade consists of multiple layers: an outer layer that shields against sunlight and minimizes heat absorption, intermediate insulating layers with air gaps to regulate heat transfer, and an inner layer that stabilizes indoor temperatures. A key feature is the "Infinity Kagome" pattern, a lattice structure that mimics natural layering to enhance airflow and heat dissipation. Additionally, built-in air channels between the layers enable passive cooling through controlled airflow, similar to ventilated façade systems in modern buildings.

4.2 Working Principle

The system operates based on passive thermal management, inspired by the silkworm cocoon's natural heat control:

- 1. Heat Absorption & Insulation:* The outer layer absorbs heat from sunlight, while intermediate layers slow down heat transfer.
- 2. Airflow Management:* Air channels between layers create a microclimate that dissipates heat efficiently. The "Infinity Kagome" pattern ensures uniform airflow and reduces hotspots.

3. *Temperature Regulation*: Interior temperatures remain stable even under extreme external conditions. Reduces reliance on active cooling systems like air conditioning.
4. *Computational Analogy*: The same principles can be applied to thermal-aware computer hardware:
 - Multi-layered cooling systems for processors.
 - Optimized airflow channels inside electronic devices.
 - Energy-efficient algorithms that mimic natural heat regulation.

4.2.1 Structure Comparison with Real-World Examples

Bio-Inspiration	Implementation in Façade	Real-World Equivalent
Silkworm cocoon layering	Multi-skin façade with layers	Double/triple-glazed windows, insulated walls
Infinity Kagome lattice	Patterned panels in façade	Honeycomb panels in architecture, heat sinks in electronics
Air circulation channels	Layered airflow paths	Ventilated façades, server room cooling ducts
Thermal regulation concept	Passive temperature control	Energy-efficient building designs, thermal-aware hardware

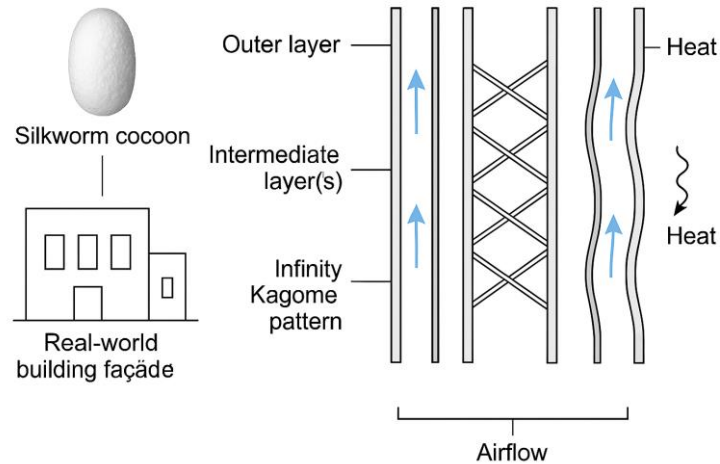


Figure 2 Inspired by Silkworm Cocoons

4.3 Outcomes & Advantages

Improves airflow and heat transfer by $\sim 27\%$, lowering energy consumption. Reduces reliance on mechanical cooling systems in buildings. Offers a direct analogy for energy-efficient computer systems, where layered design and airflow control reduce overheating and power usage. Demonstrates nature-inspired problem-solving, bridging architecture and electronics.

5. CLOUD TASK SCHEDULING INSPIRED BY LYREBIRDS

Lyrebird Falcon Optimization (LFO). This algorithm is inspired by how lyrebirds behave in nature:

"Escaping" phase $\rightarrow$ the algorithm explores different possible ways to assign tasks.

"Hiding" phase $\rightarrow$ it fine-tunes the best solutions it has found.

Together, these two steps help the algorithm:

- Efficiently assign tasks
- Balance the workload across resources
- Reduce energy usage
- Shorten the total time it takes to finish all tasks
- Use CPU and other resources more effectively

5.1 Components

It presents a cloud task scheduling approach inspired by the behaviour of lyrebirds. It explains how bio-inspired optimization can help in efficiently distributing thousands of computational tasks across multiple servers in a cloud environment. The main components include the problem definition, where the need for efficient scheduling is described, the bio-inspired model based on lyrebird behaviour, and the proposed Lyrebird Falcon Optimization (LFO) algorithm. The paper also includes a detailed comparison of LFO with traditional optimization algorithms to evaluate its performance.

5.2 Working Principle

The Lyrebird Falcon Optimization (LFO) algorithm is designed by observing how lyrebirds adapt and mimic sounds to protect themselves from predators. This natural ability is translated into two key computational phases:

- *Escaping Phase:* In this stage, the algorithm explores a wide range of possible task scheduling options. It represents the exploratory behavior of the lyrebird when escaping threats, helping the system search for diverse and effective solutions.
- *Hiding Phase:* Once potential solutions are found, the algorithm refines and fine-tunes them. This phase ensures that the most promising scheduling solutions are optimized for performance and efficiency.

Through the balance of exploration (escaping) and exploitation (hiding), LFO efficiently manages computational resources while adapting to dynamic changes in the cloud environment.

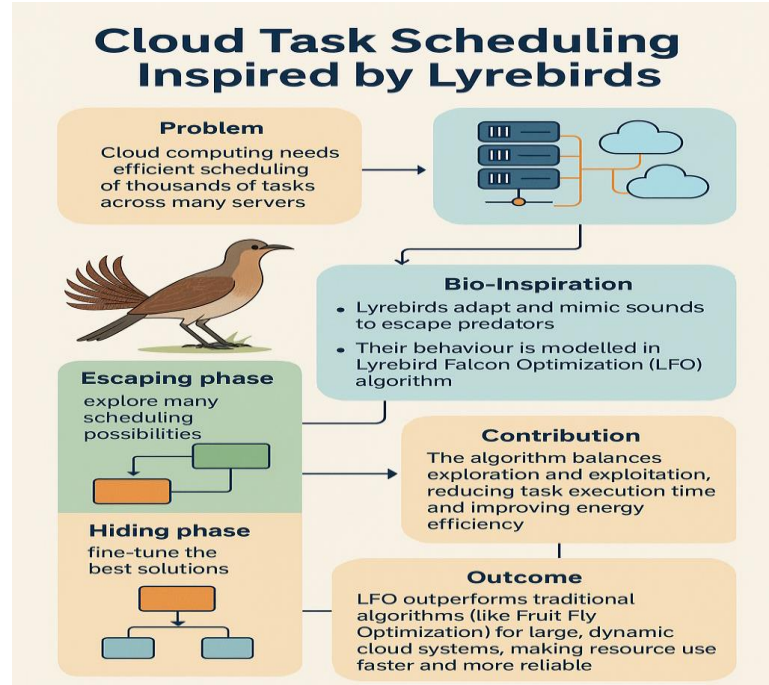


Figure 3 Inspired by Lyre Birds

5.3 Testing Process

The testing process for the LFO algorithm involved using a Cloud Sim simulation environment with workload details consisting of light workloads (1000 tasks) and heavy workloads (up to 5000 tasks). A comparison was made with the FOA (Fruit Fly Optimization Algorithm). The final summary of testing indicates that the LFO algorithm achieved lower make span, improved energy efficiency, and better utilization of CPU and resources compared to the FOA algorithm when tested in a simulated cloud environment using 1000–5000 tasks. These results demonstrate that LFO is more suitable for modern, large-scale, and dynamic cloud systems.

5.4 Outcomes and Advantages

The results show that the LFO algorithm significantly improves cloud task scheduling compared to traditional techniques like the Fruit Fly Optimization Algorithm. It achieves faster task execution times, better load balancing, and enhanced energy efficiency. By effectively balancing exploration and exploitation, LFO reduces system delays and ensures more reliable performance in large-scale and dynamic cloud systems. Overall, this bio-inspired approach provides a smarter and more adaptive solution for modern cloud computing challenges.

6. BIO-INSPIRED OPTIMIZATION FOR DISEASE DETECTION

Traditional disease detection methods rely on manual feature engineering. The Problems with these are: Redundant features → slow down training and can reduce accuracy, Time-consuming → requires lots of expert effort. Deep learning can automatically learn features, but needs large datasets and heavy computing power. Here use bio-inspired optimization algorithms to help deep learning select the most useful features automatically. These algorithms copy strategies from nature, such as:

Natural selection (Genetic Algorithms),

Social/swarm behaviour (Particle Swarm Optimization, Ant Colony Optimization).

This reduces redundancy and computing cost.

Benefits

- Better accuracy: Fewer irrelevant features → less noise → improved prediction quality.
- More efficient: Lower computational cost, works even with smaller datasets.
- More generalizable: Solutions found by these nature-inspired algorithms adapt well to different kinds of medical data.

6.1 Structure

This focuses on the use of bio-inspired optimization techniques to improve disease detection through deep learning. It explains how nature-inspired algorithms can enhance the process of analyzing medical images by automatically selecting important features. The structure of the paper includes the problem definition, the introduction of bio-inspired algorithms, the proposed feature selection method, and an evaluation of the approach. The

main idea is that selecting the right features manually is both difficult and time-consuming, and bio-inspired algorithms can make this process faster and more accurate.

6.2 Working Principle

Deep Learning and Optimization: This approach combines deep learning with bio-inspired optimization algorithms to enhance disease detection from medical images.

Nature-Inspired Methods: Algorithms like Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and Ant Colony Optimization (ACO) mimic natural processes such as evolution, group movement, and ant foraging to find the best solutions.

Feature Selection: These algorithms automatically select the most relevant features from medical images, removing redundant data and focusing on key patterns.

Better Model Training: By selecting useful features, the method improves training speed, accuracy, and the model's ability to learn effectively even from limited or complex medical data.

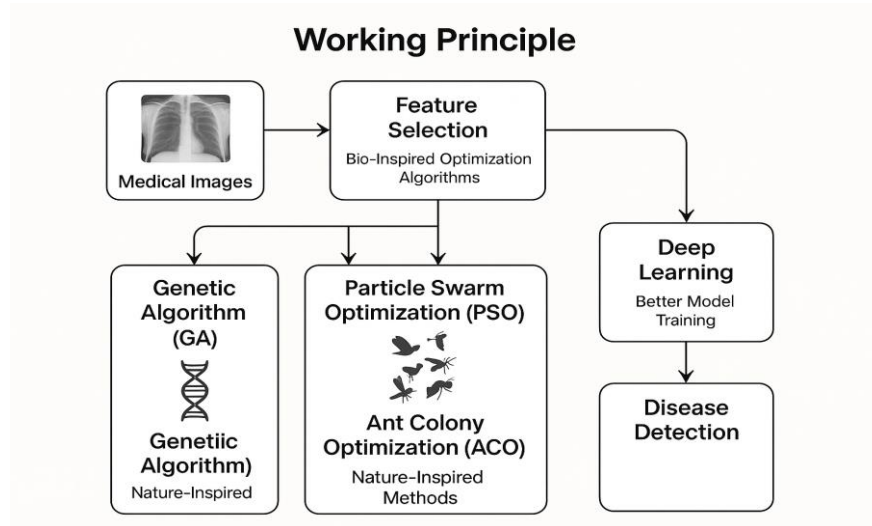


Figure 4 Working Principle of Disease Detection

6.3 Outcomes

The results of this study show that bio-inspired optimization greatly enhances both the efficiency and accuracy of disease detection systems. The optimized models perform better because they are trained on more relevant features and require less time for computation. This approach also reduces the need for manual feature engineering, saving valuable time for researchers and medical professionals. Moreover, it performs well even when dealing

with high-dimensional or limited datasets, which are common in medical applications. Over all, this bio-inspired approach provides a smart, adaptive, and efficient solution for disease detection. It allows deep learning models to work faster, use fewer resources, and deliver more accurate diagnostic results - making it a valuable advancement in modern medical image analysis.

7. INSECT-INSPIRED BIOLOGICAL SENSORS & ROBOTICS

This discusses the development of insect-inspired biological sensors and robotics, highlighting how designs from nature can enhance modern technology. It focuses on creating lightweight, highly efficient, and sensitive systems for real-world applications. The paper begins with the problem of efficiency and sensitivity in current robotic systems, followed by an explanation of bio-inspired models derived from insects. It also presents the proposed designs and contributions, comparing them to traditional human-made systems to demonstrate their effectiveness and adaptability.

7.1 Mimicking - Working Principle

The working principle of insect-inspired technology is based on mimicking the sensory and movement mechanisms found in insects. The compound eyes of flies inspire wide-angle, high-speed cameras capable of detecting motion quickly from multiple directions. The antennae of moths and bees serve as models for chemical and gas sensors that can detect tiny concentrations of substances in the air. Similarly, the locomotion of ants and beetles inspires robots that can move efficiently across uneven or complex terrain. By replicating these biological mechanisms, engineers design robotic systems that combine agility, sensitivity, and adaptability allowing them to operate effectively in challenging environments.

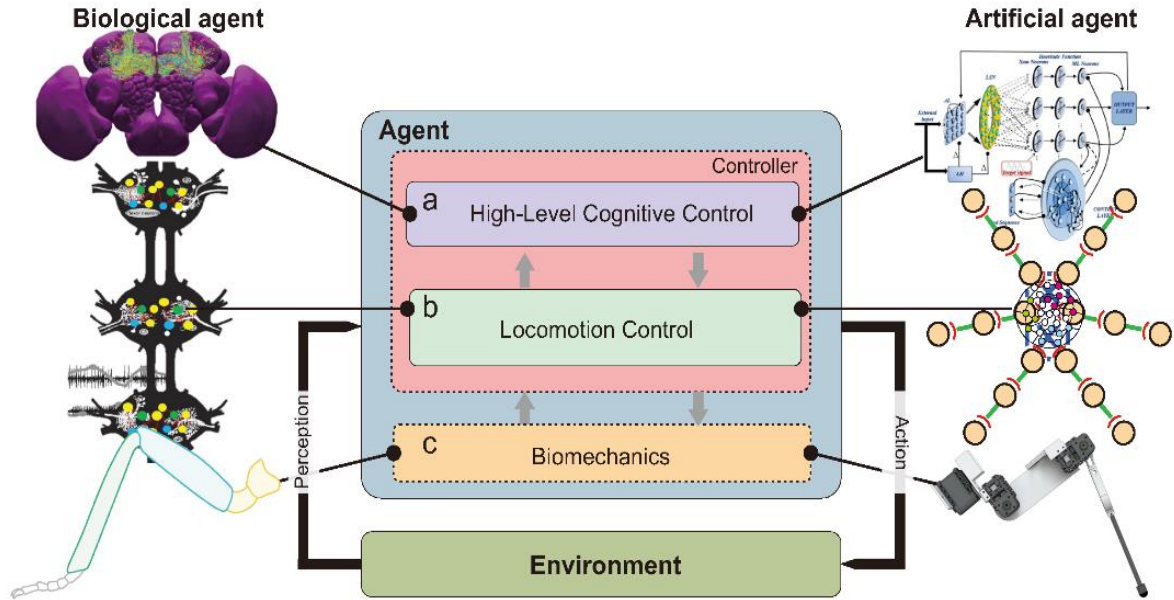


Figure 5 Mimicking behavior

It shows that insect-inspired designs have led to lighter, cheaper, and more efficient sensors and robots. These systems outperform many conventional technologies in terms of mobility, sensitivity, and cost-effectiveness. They are increasingly being used in environmental monitoring, search and rescue missions, and micro-robotics. Over all, this bio-inspired approach demonstrates how studying nature's tiny creatures can lead to major technological innovations, bridging biology and engineering to create smarter and more responsive machines.

8. BIO-INSPIRED OPTIMIZATION-BASED UAV PATH PLANNING

- It introduces the bio-inspired models that replicate swarm behavior, bird flocking, and ant colony foraging, forming the foundation of the proposed algorithm. The system's structure includes environmental mapping, path optimization, and real-time route adjustment modules. These algorithms mimic natural intelligence swarm cooperation, self-organization, survival of the fittest.
- Nature's methods of finding optimal solutions in complex, changing environments are turned into mathematical models for computing.

8.1 Nature Inspired - Working Principle

Nature-Inspired Behavior: The working principle of this approach lies in mimicking natural collective behaviors to optimize UAV flight paths. Just like birds coordinating in flocks or ants finding the shortest route to food, the algorithm allows drones to explore multiple possible paths and identify the most efficient one.

Swarm Intelligence Techniques: Using principles of swarm intelligence, the system maintains a balance between exploration (searching for new and better routes) and exploitation (refining the best route found). This ensures that the UAV doesn't get stuck with suboptimal paths and continues to improve its route dynamically.

Real-Time Adaptation: When obstacles or environmental changes occur, the algorithm quickly adapts by recalculating new paths in real time. This enables the UAV to maintain safe, collision-free movement even in unpredictable conditions.

Efficient Navigation: By combining nature-inspired intelligence with real-time optimization, the UAV achieves efficient, smooth, and energy-saving flight performance across dynamic and complex environments.

8.2 In UAV Path Planning

- *Goal:* Find the shortest, safest, energy-efficient route while avoiding obstacles and reacting to changing conditions.
- *Challenge:* The search space is huge and has many variables; traditional math optimization is too slow or gets stuck.
- *Bio-inspired solution:* Nature-mimicking algorithms explore many possibilities in parallel, adapt quickly, and find a near-optimal path efficiently.

8.3 Optimization methods that copy strategies from living systems

Nature's Behavior	Algorithms	How it is used in UAV's
Bird Flocking/fish schooling	Particle Swarm Optimization (PSO)	Drones are treated like "particles" that share information like a flock,

		gradually moving toward the best path.
Ants laying pheromone trails	Ant Colony Optimization (ACO)	Virtual “pheromones” mark good routes; the UAV follows the strongest trail to find the shortest, safest path.
Evolution and natural selection	Genetic Algorithm (GA)	Possible flight paths are like a population
Bees finding food	Artificial Bee Colony	UAVs mimic how bees scout and share the best food sources to locate optimal routes.

This shows bio-inspired computing by taking problem-solving strategies from biology (like how ants, bees, birds, or evolution work) and turning them into computational optimization algorithms that guide drones (UAVs) to plan their flight paths.

8.4 Advantages

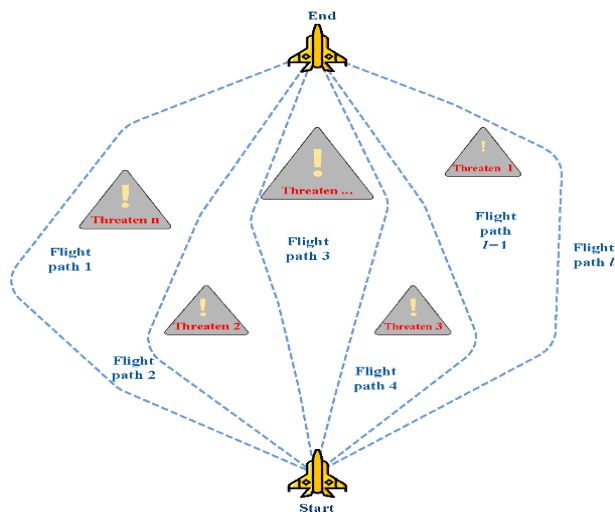


Figure 6 UAV Path Finding

The results of this method show that bio-inspired optimization significantly improves UAV performance in path planning. The drones are able to find shorter and safer routes while minimizing flight time and energy consumption. The adaptive nature of the algorithm also reduces the risk of collisions and ensures smoother movement in unpredictable environments.

Overall, this bio-inspired path-planning approach offers a fast, reliable, and energy-efficient solution for modern UAV systems, making them more effective for tasks like surveillance, disaster response, and environmental monitoring.

9. LITERATURE REVIEW

- **Optimization Algorithms:** The Genetic Algorithm (GA) mimics natural selection to find near-optimal solutions for complex problems like scheduling and route optimization. Swarm intelligence methods, such as Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO), are used in systems like deep learning for disease detection to automatically select the most relevant features, improving accuracy and reducing computation costs.
- **Bio-Inspired Engineering:** Designs leverage natural stability and efficiency. Quantum Computing Inspired by Chloroplasts proposes a qubit layout that mimics the thylakoid structure to operate stably at room temperature, lowering energy costs. Similarly, Multi-Skin Facades based on the silkworm cocoon's structure offer passive thermal management for buildings, a principle applicable to cooling computer hardware.
- **Adaptive Systems:** The Lyrebird Falcon Optimization (LFO) algorithm, which models the lyrebird's escaping and hiding behaviours, efficiently balances task assignment and workload in Cloud Task Scheduling. In UAV Path Planning, algorithms based on bird flocking and ant foraging enable drones to find optimal, energy-efficient paths while adapting to dynamic obstacles.

10. CONCLUSION AND FUTURE DIRECTIONS

When we let nature teach us how to compute, we build systems that are not only smarter but also inherently more sustainable. This proposal for a Hybrid Bio-Inspired Framework for Smart Environmental Monitoring extends that vision showing how evolutionary algorithms and insect olfaction models can work together to protect our environment and health. *Hybrid Bio-Inspired Framework for Smart Environmental Monitoring Proposal:* The project aims to build a small UAV swarm with insect-inspired chemical sensors and use a Genetic Algorithm (GA) to optimize flight paths for efficient air quality mapping in urban areas. By combining bio-mimetic sensing with GA-based optimization, the swarm will monitor air quality in real-time using minimal energy while maximizing coverage and detection accuracy. The framework leverages insect olfaction for sensitive pollutant detection and GA for adaptive path planning, enabling low-cost, energy-efficient air quality mapping with potential applications in smart-city planning and public health alerts.

Key Components:

1. *Insect-Inspired Chemical Sensors:* These sensors mimic the olfactory systems of insects, such as moths and bees, to detect minute chemical traces in the air. They will be mounted on each UAV to measure pollutant concentrations (e.g., NO₂, CO, particulate matter).
2. *Genetic Algorithm (GA) Optimization:* The GA will manage the swarm's collective paths, optimizing for coverage area, energy use, and sensor sampling density. The algorithm will adapt to changing environmental conditions, such as wind and pollution hotspots.
3. *UAV Swarm:* A group of UAVs will work together to map air quality in a designated area, transmitting data to a ground station in real-time.

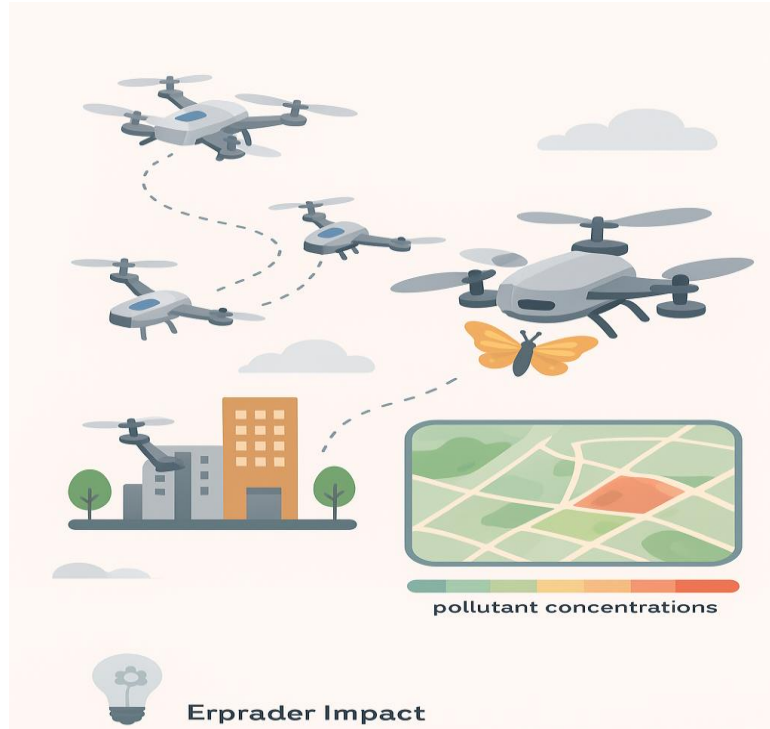


Figure 7 UAV Swarm for Smart Air Quality Monitoring with Bio-Inspired Computing

Expected Outcomes:

1. *Improved Air Quality Mapping:* The system will provide detailed maps of air quality, to enable better understanding of pollution dynamics and informing policy decisions.
2. *Adaptive and Scalable:* The system will adapt to changing environmental conditions and can be scaled up or down depending on requirements.
3. *Potential Applications:* The system has potential applications in smart-city planning, public health alerts, and environmental monitoring.

Broader Impact:

The proposal demonstrates the potential of bio-inspired computing in solving complex real-world problems. By leveraging biological principles and evolutionary optimization, the system offers a novel approach to environmental monitoring, with potential applications in various fields.

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12. APPENDIX

S. No	Title, Author and Year	Research Area	Methodology/ Algorithm Used	Contributions in the Paper	Future Work
1	T. Mishra, P. Desai, and A. Rao, 2024	Bio-Inspired Quantum Computing	Exciton Coherence Inspiration and Thylakoid-Inspired Qubit Layout (from chloroplasts)	Proposes stable, energy-efficient qubit layouts that inherently reduce the need for extreme cooling and external error correction.	Leads the way toward scalable, efficient, and practical deployment of quantum processors under less restrictive environmental conditions.
2	M. H Vajari et al., 2025	Bio-Inspired Design/Thermal Management	Multi-Skin Facade inspired by Silkworm Cocoon layering, using the Infinity Kagome pattern.	Optimizes bio-inspired macro-structures for enhanced thermal efficiency in buildings, improving airflow and heat transfer by ~27%.	Principles are applicable to layered cooling systems and optimized airflow channels for thermal management in computing hardware
3	G. Maddali, 2025	Cloud Task Scheduling	Lyrebird Falcon Optimization (LFO) algorithm, featuring Escaping (exploration) and Hiding.	Achieves faster task execution, better load balancing, and enhanced energy efficiency compared to traditional algorithms like FOA.	Provides a smarter and more adaptive solution suitable for modern large-scale, dynamic cloud systems and resource management.

S. No	Title, Author and Year	Research Area	Methodology/ Algorithm Used	Contributions in the Paper	Future Work
4	Ashwini et al., 2025	Bio-Inspired Disease Detection	Genetic Algorithm (GA), PSO, and ACO used for Automatic Feature Selection	Improves accuracy, lowers computational cost, and enhances the generalizability of Deep Learning models for disease detection.	The approach is effective in dealing with high-dimensional or limited datasets, which are common challenges in medical applications.
5	E. Abbasi, 2025	Insect-Inspired Robotics & Sensors	Mimicking the sensory (antennae, compound eyes) and locomotion mechanisms of insects (e.g., moths, flies, ants).	Led to lighter, cheaper, more efficient sensors and robots, outperforming conventional technology in terms of mobility and sensitivity.	Applications in environmental monitoring, search and rescue missions, and the development of high-speed micro-robotics.
6	S. Poudel et al., 2023	UAV Path Planning	Nature-mimicking algorithms including PSO, ACO, and GA for route optimization.	Enables UAVs to find shorter, safer, and energy-efficient routes by reducing collision risk and ensuring smoother movement in unpredictable environments	Facilitates real-time route adjustment and adaptive navigation across dynamic, complex environments for tasks like surveillance and disaster response.

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7	(Foundational Concept)	Genetic Algorithm (GA)	Simulates Natural Selection and Evolutionary Principles	Provides near-optimal solutions for complex problems with large search spaces where exact solutions are hard to compute (e.g., routing, scheduling).	This fundamental algorithm is often hybridized and applied across all optimization domains, including the proposed future work.
8	(Foundational Concept)	Particle Swarm Optimization (PSO)	Models Bird Flocking or Fish Schooling, where particles (drones) share information	Allows UAVs to collaboratively explore and converge on the best path efficiently, balancing exploration and exploitation. (Used in UAVs)	A key algorithm enabling real-time route adjustment and efficient navigation in dynamic environments (e.g., used in UAVs).
9	(Foundational Concept)	Ant Colony Optimization (ACO)	Models Ant Foraging behavior, using Virtual Pheromones to mark the most successful routes.	Used to guide systems (like UAVs or Deep Learning feature selectors) to follow the strongest virtual trail to find the shortest/optimal solution.	Crucial for dynamic path planning, allowing quick recalculation when obstacles or environmental changes occur (e.g., used in UAVs).
10	(Proposed Framework)	Future Environmental Monitoring	Integration of Insect-Inspired Chemical Sensors and Genetic Algorithm (GA) for swarm control.	Proposal: Provides a novel framework for adaptive, real-time, low-cost air quality mapping using a UAV swarm.	Applications in smart-city planning, public health alerts, and providing detailed maps of air pollution dynamics.